

Qualification Pack



Mineral Processing Technician

QP Code: MIN/Q1804

Version: 1.0

NSQF Level: 4.0

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MIN/Q1804: Mineral Processing Technician

Brief Job Description

A Mineral Processing Technician efficiently processes mineral ores and employs specialized techniques, including magnetic and electrostatic separation, to extract valuable minerals while adhering to quality and safety standards in the industry.

Personal Attributes

This job requires the individual to concentrate on the task at hand and complete it efficiently and effectively. The individual should also be physically agile, strong, have good eye sight and not suffer from colour-blindness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N4109: Carry out rare earth material ore processing](#)
2. [MIN/N4110: Convey the material from mine for processing](#)
3. [MIN/N4111: Operate the spiral separator and density separator](#)
4. [MIN/N4112: Operate the drier and blower](#)
5. [MIN/N4113: Operate the electrostatic separators, magnetic separators, screens, air table](#)
6. [MIN/N4114: Carry out wet table separation, flotation separation, dewatering and screening](#)
7. [MIN/N4115: Carry out general maintenance of ore processing equipment](#)
8. [MIN/N4116: Perform water treatment and management of subgrade ore](#)
9. [MIN/N1810: Follow the applicable health, safety and environmental guidelines \(Placer Mining\)](#)
10. [MIN/N1811: Work collaboratively and effectively in a team](#)
11. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Mining
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Sub-Sector	Mining Operation
Occupation	Placer Mining
Country	India
NSQF Level	4.0
Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 3121.9900
Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of the 3-year diploma after 10 (in Mining/ Chemical/ Metallurgy/ Material Science) OR 10th grade pass (with two years of NTC/ NAC/ CITS or equivalent (in Mechanical/ Electrical)) OR 11th grade pass (with 1 year relevant experience) OR 10th grade pass (with 2 years relevant experience) OR 8th grade pass with 2 year NTC plus 1 year NAC plus 1 year CITS (in Mechanical/ Electrical) OR Previous relevant Qualification of NSQF Level (3.0) (with 3 years relevant experience) OR Previous relevant Qualification of NSQF Level (3.5)(with 1.5 year relevant experience)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	14/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-04-MG-02173-2024-V1-SCMS
NQR Version	1.0

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Remarks:

450 Hours

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MIN/N4109: Carry out rare earth material ore processing

Description

This unit is about understanding the job requirement, equipment and processes for ore processing as per the specified standards

Scope

The scope covers the following :

- Ore processing procedure, related equipment and parameters

Elements and Performance Criteria

Ore processing procedure, related equipment and parameters

To be competent, the user/individual on the job must be able to:

- PC1.** carry out ore processing activity following the process manuals/ work instructions and Standard Operating Procedures (SOP)
- PC2.** use the appropriate reagents in the recommended quantities in ore processing, applying effective reagents handling techniques
- PC3.** monitor and record the process parameters during ore processing as per the standard procedure
- PC4.** use the appropriate hand and power tools, such as pneumatic and hydraulic tools at different stages of ore processing
- PC5.** use the appropriate equipment as per the manufacturer's instructions, such as conveyor, compressor, mobile equipment and lifting equipment, at various stages of ore processing
- PC6.** prepare for the tools and equipment-related emergencies, such as malfunction and breakdown
- PC7.** follow the recommended emergency procedure during the malfunctioning or breakdown of the tools and equipment

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of minerals extracted from ore mines
- KU2.** different ore processing processes and appropriate equipment/machinery used in the mine
- KU3.** the statutory requirements and safety regulations for ore processing operations
- KU4.** the importance of discipline and punctuality in ore processing operations
- KU5.** the benefits of following 5s principles in ore processing
- KU6.** the appropriate safety precautions for working in the vicinity of ore processing machinery
- KU7.** the duties of workmen and applicable working hours in the ore processing plant
- KU8.** the provision of wages, working hours and accident compensation as per the Mines act
- KU9.** the impact of violation of safety procedures

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- KU10.** the appropriate process parameters, such as belt speed, voltage, oil temperature, dryer temperature and their operational values
- KU11.** different types of reagents, such as acids, alkalis, carbon, etc., and their mixing proportions
- KU12.** the relevant hazards and safety aspects of ore processing activities, e.g. the handling reagents and Usage of relevant PPEs.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write information documents to internal departments/ teams or enter the information in online ERP systems under the guidance of the supervisor
- GS2.** read equipment manuals and process documents for the better understanding of equipment and processes
- GS3.** discuss task lists, schedules and activities with the supervisor
- GS4.** communicate effectively with team members
- GS5.** plan and organize work, scheduling tasks for the smooth running of operations
- GS6.** make appropriate decisions concerning the area of work, following the authority matrix defined by the organization
- GS7.** follow the given instructions and work on the areas of improvement
- GS8.** carry out the assigned tasks within the applicable timelines, maintaining the applicable quality standards
- GS9.** identify and discuss appropriate solutions to work-related problems with the supervisor

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ore processing procedure, related equipment and parameters</i>	30	50	-	20
PC1. carry out ore processing activity following the process manuals/ work instructions and Standard Operating Procedures (SOP)	6	10	-	4
PC2. use the appropriate reagents in the recommended quantities in ore processing, applying effective reagents handling techniques	5	8	-	3
PC3. monitor and record the process parameters during ore processing as per the standard procedure	6	7	-	2
PC4. use the appropriate hand and power tools, such as pneumatic and hydraulic tools at different stages of ore processing	3	7	-	2
PC5. use the appropriate equipment as per the manufacturer's instructions, such as conveyor, compressor, mobile equipment and lifting equipment, at various stages of ore processing	3	7	-	3
PC6. prepare for the tools and equipment-related emergencies, such as malfunction and breakdown	5	7	-	3
PC7. follow the recommended emergency procedure during the malfunctioning or breakdown of the tools and equipment	2	4	-	3
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4109
NOS Name	Carry out rare earth material ore processing
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N4110: Convey the material from mine for processing

Description

This unit is about conveying the ore to the processing plant through slurry pipeline systems and conveyors.

Scope

The scope covers the following :

- Operation of the conveyors
- Operation of the feeders
- Operation of the pumps and pipelines

Elements and Performance Criteria

Operation of the conveyors

To be competent, the user/individual on the job must be able to:

- PC1.** check the slurry pumping system, layout of the pipeline for the correct functioning
- PC2.** check the appropriate parameters, such as feed rate, pulp density etc., for components like pumps and motors
- PC3.** check the layout and components of belt conveyor system including its principal componentse.g., through scrapers, conveyor belt, roller chains and tensioners for the correct functioning
- PC4.** test the operations of mechanical and electrical driving units including the safety devices
- PC5.** convey the ore to the processing unit, maintaining appropriate parameters, such as feed/material rate, alignment of conveyor belts, spillage amount and rate, conveyor jams, etc
- PC6.** inspect for potential issues, such as unusual noises, blockages, obstruction, etc
- PC7.** undertake the lubrication of pumps, motors, conveyors and electromagnet
- PC8.** inspect the hoppers and bins, chutes, guardrails deployed to control stockpile levels, stability of stockpiles, clearing of the obstruction for the correct installation

Operation of the feeders

To be competent, the user/individual on the job must be able to:

- PC9.** check the layout, functioning and components of feeders, including chains, pans, bearings, rollers, shafts, feed/discharge chute, drive mechanisms, lubrication systems, etc. controls, drive belts, hangers/cables, etc
- PC10.** feed ore to the various processing equipment maintaining the relevant parameters, e.g. feed rate, voltage, Revolution per Minute (RPM) etc., as per the applicable standards and recommendations
- PC11.** follow the appropriate preventive or corrective measures for safety hazards, such as spillage, abnormal noises/ odour, lubrication, leaks, etc

Operation of the pumps and pipelines

To be competent, the user/individual on the job must be able to:

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- PC12.** convey ore to the processing unit through pipelines
- PC13.** check the Pitch Circle Diameter (PCD) of pipelines, flanges and their fittings, and different types of fasteners used for pipelines for the correct installation
- PC14.** identify faults in pumps and carry out appropriate troubleshooting, making suitable adjustments to the machinery where necessary to improve the performance
- PC15.** follow the appropriate preventive and corrective measures for safety hazards, such as spillage, abnormal noises/odours, leaks, etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the slurry pumping system, layout of the pipeline and functioning of components, such as pumps, motors, and parameters, e.g. feed rate, pulp density, etc
- KU2.** the layout and components of belt conveyor system, e.g. through scrapers, conveyor belt, clips, roller chains, guards, tensioners, feed and discharge chutes, head, tail and take-up pulleys, bearings, braking system, controls such as switches, trip cords, inter locks, alarms, etc.
- KU3.** the operations of mechanical and electrical driving units including safety devices in the belt conveyor system
- KU4.** different types of mineral processing processes and the equipment used for the purpose
- KU5.** different types of pumps and conveyors, such as belt types, screw types, pneumatic types, etc
- KU6.** the maintenance of pipeline systems, pumps, motor, conveyor and principal conveyors components, such as screw, screw casing/trough scrapers, conveyor belt, splices or clips, drive mechanisms, roller chains/belts, buckets, guards, tensioners, feed and discharge chutes, head, tail and take-up pulleys, idlers, bearings, braking system, controls (such as switches, trip cords, side travel switches, interlocks, crossovers, audible alarms, magnet, metal detectors, and lubricating systems
- KU7.** different types of feeders, such as pan feeder, screw feeder, belt feeder, pneumatic feeder, tube feeder, vibratory feeder, tripper, rotary valve feeder, drag feeder, etc
- KU8.** the principal components of feeders, such as, chains, pans, bearings (support, hangers, pillow block bearings), sprockets, rollers, roller mounts, return rollers, shafts, centre support rail, feed/discharge chute, drive mechanisms, gear reducer, lubrication systems, guards, controls, scraper, fire extinguishing equipment, drive belts, hangers/cables, etc
- KU9.** the safety instructions for conveying systems (pull chord, label switch, fire alarm etc
- KU10.** the belt mechanisms/ systems (belt fastening, belt jointing, etc

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** interpret the relevant sketches and engineering drawings, and make note of observations, if any
- GS2.** read equipment manuals and process documents for the better understanding of equipment and processes



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- GS3.** prepare draft drawings for the final output product
- GS4.** enter the relevant information in the online ERP system under guidance of the supervisor
- GS5.** discuss task lists, schedules and activities with the supervisor
- GS6.** communicate effectively with team members
- GS7.** plan and organize work, scheduling tasks for smooth running of operations
- GS8.** organize all process/equipment manuals so that sorting/accessing information is easy
- GS9.** make appropriate decisions concerning the area of work, following the authority matrix defined by the organization
- GS10.** follow the given instructions and work on the areas of improvement
- GS11.** carry out the assigned tasks within the applicable timelines, maintaining the applicable quality standards
- GS12.** identify and discuss appropriate solutions to work-related problems with the supervisor
- GS13.** take initiatives in driving small projects with the supervisor, e.g. operation improvement, training of helpers and assistant operators, etc

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operation of the conveyors</i>	20	27	-	10
PC1. check the slurry pumping system, layout of the pipeline for the correct functioning	3	4	-	1
PC2. check the appropriate parameters, such as feed rate, pulp density etc., for components like pumps and motors	3	4	-	2
PC3. check the layout and components of belt conveyor system including its principal componentse.g., through scrapers, conveyor belt, roller chains and tensioners for the correct functioning	3	4	-	2
PC4. test the operations of mechanical and electrical driving units including the safety devices	3	3	-	1
PC5. convey the ore to the processing unit, maintaining appropriate parameters, such as feed/material rate, alignment of conveyor belts, spillage amount and rate, conveyor jams, etc	2	3	-	1
PC6. inspect for potential issues, such as unusual noises, blockages, obstruction, etc	2	3	-	1
PC7. undertake the lubrication of pumps, motors, conveyors and electromagnet	2	3	-	1
PC8. inspect the hoppers and bins, chutes, guardrails deployed to control stockpile levels, stability of stockpiles, clearing of the obstruction for the correct installation	2	3	-	1
<i>Operation of the feeders</i>	5	11	-	4
PC9. check the layout, functioning and components of feeders, including chains, pans, bearings, rollers, shafts, feed/discharge chute, drive mechanisms, lubrication systems, etc. controls, drive belts, hangers/cables, etc	2	4	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. feed ore to the various processing equipment maintaining the relevant parameters, e.g. feed rate, voltage, Revolution per Minute (RPM) etc., as per the applicable standards and recommendations	2	4	-	1
PC11. follow the appropriate preventive or corrective measures for safety hazards, such as spillage, abnormal noises/ odour, lubrication, leaks, etc	1	3	-	1
<i>Operation of the pumps and pipelines</i>	5	12	-	6
PC12. convey ore to the processing unit through pipelines	1	2	-	1
PC13. check the Pitch Circle Diameter (PCD) of pipelines, flanges and their fittings, and different types of fasteners used for pipelines for the correct installation	2	3	-	2
PC14. identify faults in pumps and carry out appropriate troubleshooting, making suitable adjustments to the machinery where necessary to improve the performance	1	4	-	2
PC15. follow the appropriate preventive and corrective measures for safety hazards, such as spillage, abnormal noises/odours, leaks, etc	1	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4110
NOS Name	Convey the material from mine for processing
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N4111: Operate the spiral separator and density separator

Description

This unit is about spiral and density separation process and operation.

Scope

The scope covers the following :

- Spiral separator operation
- Density separator operation

Elements and Performance Criteria

Spiral separator operation

To be competent, the user/individual on the job must be able to:

- PC1.** check different operational parameters, such as tank level, alarms, pumps, monitor-pulp density, feed rate, splitter position, etc., at appropriate intervals following the applicable checklists and given instructions
- PC2.** check for the recommended feed rate for spiral ore separation process
- PC3.** check the pulp density is as per the process requirement
- PC4.** conduct the separation process following the applicable standards to extract heavy and lighter minerals
- PC5.** check and monitor the threshold limit values to ensure they are as recommended
- PC6.** assist the supervisor in identifying relevant hazards by checking and reporting the work site conditions
- PC7.** mitigate the identified hazards, such as spillage, blockages and obstructions by using ropes/barriers, signboards, etc.

Density separator operation

To be competent, the user/individual on the job must be able to:

- PC8.** check the functioning of different types of density separators, e.g. hydro-sizer and flotex density separator
- PC9.** monitor the threshold limit values of the density separator

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of density separators, such as hydro-sizer, flotex density separator and its working principle
- KU2.** different types of heavy minerals separation processes
- KU3.** the functioning of various alarms, pumps, screens, valve, conveyor etc., relevant to ore separation equipment
- KU4.** the use of reagents as per the process requirements

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- KU5.** the functioning of hydraulic systems
- KU6.** the safety procedures for operating spiral separator
- KU7.** the troubleshooting of spiral and density separators
- KU8.** different types of filtration systems and components including stock tanks, pumping systems, sampling systems, drive systems, lubrication systems, hydraulic systems, pneumatic systems, vacuum systems, filter cloth, etc
- KU9.** the basic repair and maintenance of gearbox, bearings, pump, hydraulic systems, motors, valves etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read the internal instructions, equipment specifications, and parameters on different monitoring panels
- GS2.** document information from the sketches, engineering drawings, and various process parameters in the required reporting format
- GS3.** discuss task lists, schedules and activities with the supervisor and assist in scheduling tasks for subordinates, e.g. helper and assistant operator
- GS4.** analyze the data available about the site
- GS5.** communicate effectively with the team members and assist the subordinates in completing their assigned tasks
- GS6.** listen attentively with full attention and comprehend the information being given
- GS7.** complete the assigned tasks within the applicable time limits with minimum supervision and maintain quality
- GS8.** identify work-related problems and appropriate solutions to them, discussing possible solutions with the supervisor
- GS9.** make appropriate decisions in case of emergencies, following the Authority matrix defined by the organization

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Spiral separator operation</i>	22	40	-	16
PC1. check different operational parameters, such as tank level, alarms, pumps, monitor-pulp density, feed rate, splitter position, etc., at appropriate intervals following the applicable checklists and given instructions	4	6	-	3
PC2. check for the recommended feed rate for spiral ore separation process	4	6	-	3
PC3. check the pulp density is as per the process requirement	4	6	-	3
PC4. conduct the separation process following the applicable standards to extract heavy and lighter minerals	2	5	-	2
PC5. check and monitor the threshold limit values to ensure they are as recommended	4	6	-	3
PC6. assist the supervisor in identifying relevant hazards by checking and reporting the work site conditions	2	6	-	1
PC7. mitigate the identified hazards, such as spillage, blockages and obstructions by using ropes/barriers, signboards, etc.	2	5	-	1
<i>Density separator operation</i>	8	10	-	4
PC8. check the functioning of different types of density separators, e.g. hydro-sizer and flotex density separator	6	6	-	3
PC9. monitor the threshold limit values of the density separator	2	4	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4111
NOS Name	Operate the spiral separator and density separator
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N4112: Operate the drier and blower

Description

This unit is about operating the drier and blower during ore processing.

Scope

The scope covers the following :

- Drier and blower operation

Elements and Performance Criteria

Drier and blower operation

To be competent, the user/individual on the job must be able to:

- PC1.** use different types of driers, e.g. rotary drier, fluidized bed drier, shaft drier; and blowers as per the requirement
- PC2.** inspect the drier components, such as feed gate, discharge point, guards, feed line, launders, skirting, fuel line, burner, fuel pumps, blowers, electrical control and temperature control for wear and tear and correct functioning
- PC3.** carry out sampling testing as required by adjusting the fuel rate, temperature, moisture content, and dust control by using dust collectors
- PC4.** follow the appropriate preventive or corrective measures for safety hazards, such as spillage, abnormal noises/ odour, lubrication, leaks, etc

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of drier, such as rotary drier, fluidized bed drier, shaft drier, etc
- KU2.** the appropriate safety precautions to be taken in the vicinity of a drier
- KU3.** different components of a drier, such as drive motors, blower fan, cooling motor, bearings, different types of gear, shell, liners, lubrication system, moisture control, alarms, etc
- KU4.** the plant equipment isolation/lock-off policies and procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note measurements, equipment panel readings for various process parameters in the required reporting formats
- GS2.** Read and interpret engineering drawing, sketches, and symbols and measurements used in them
- GS3.** communicate effectively with the team members



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- GS4.** organize all process/equipment manuals so that sorting/accessing information is easy
- GS5.** detect problems in day-to-day tasks and identify appropriate solutions through discussion with the supervisor
- GS6.** maintain discipline and punctuality at work
- GS7.** complete the assigned tasks within the applicable time limits with minimum supervision and maintain quality
- GS8.** make appropriate decisions in case of emergencies, following the authority matrix defined by the organization

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Drier and blower operation</i>	30	50	-	20
PC1. use different types of driers, e.g. rotary drier, fluidized bed drier, shaft drier; and blowers as per the requirement	7	10	-	5
PC2. inspect the drier components, such as feed gate, discharge point, guards, feed line, launders, skirting, fuel line, burner, fuel pumps, blowers, electrical control and temperature control for wear and tear and correct functioning	8	15	-	5
PC3. carry out sampling testing as required by adjusting the fuel rate, temperature, moisture content, and dust control by using dust collectors	8	15	-	5
PC4. follow the appropriate preventive or corrective measures for safety hazards, such as spillage, abnormal noises/ odour, lubrication, leaks, etc	7	10	-	5
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4112
NOS Name	Operate the drier and blower
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N4113: Operate the electrostatic separators, magnetic separators, screens, air table

Description

This unit is about segregating minerals using electrostatic and magnetic separators, screens, and air table

Scope

The scope covers the following :

- Operate the electrostatic separators
- Operate the magnetic separator and the screen
- Operate the air table

Elements and Performance Criteria

Operate the electrostatic separators

To be competent, the user/individual on the job must be able to:

- PC1.** check the layout and components of electrostatic separators, e.g. drive motor, motor cooling fan, drive shaft and bearings, rollers, lubrication system, power supply unit, etc
- PC2.** conduct electrostatic separation operation maintaining the appropriate parameters, such as load limitation, roll speed, electrode position, voltage, feed material temperature, etc
- PC3.** check and ensure the feed and discharge points are free from obstructions and blockages
- PC4.** follow the appropriate preventive and corrective measures, e.g. earthing, door interlock, cleanup spills, leaks, etc., for the identified hazards

Operate the magnetic separator and the screen

To be competent, the user/individual on the job must be able to:

- PC5.** use different types of magnetic separators as per the manufacturer's instructions
- PC6.** check for the correct layout and functioning of magnetic separation unit components, e.g. drive mechanism pulley, power supply unit, roller, bearings, splitters, feed arrangements, feed belt, etc.
- PC7.** check for potential issues during the operation of magnetic separator, such as unusual noises, blockage, obstructions, magnetic intensity, leaks, voltage supply, etc
- PC8.** check for blockages at the recommended intervals and clear them appropriately
- PC9.** conduct magnetic separation operation by controlling the feed/ material by adjusting opening of gates, chutes and gaps, adjusting the speed of feeder, and controlling dust by using dust collectors
- PC10.** remove the excess spillage caused during magnetic separation
- PC11.** carry out processing for separating feed/ material, working around the screens, including control of dust, control of feed/material flow rate
- PC12.** follow the appropriate preventive and corrective measures for safety hazards, such as spillage, abnormal noises/odour, lubrication, leaks, etc

Operate the air table

To be competent, the user/individual on the job must be able to:

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PC13. check the moisture of feed material

PC14. check the relevant parameters, e.g. inclination of the air table, table cloth, hot air blower, and splitter position

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of electrostatic separators, e.g. high tension separator and electrostatic plate separator

KU2. different components of electrostatic separators, such as drive motors, electrode, power supply systems, drive shafts and bearings, brushes, lubrication systems, etc.

KU3. the appropriate safety precautions to be taken while working in the vicinity of electrostatic and magnetic separators

KU4. different types of alarms, pumps, screens, monitors etc., relevant electrostatic separators

KU5. the plant equipment isolation/lock-off policies and procedures

KU6. different types of heavy mineral processes and associated equipment

KU7. different types of magnetic separator principal components, such as gates, chutes and gaps, feeder, conveyor, dust collectors, etc

KU8. different types of screening equipment, such as drive mechanism, balance wheel, rocker arms, conveyor belt, etc.

KU9. the basic repair and maintenance of electrostatic separators

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read equipment manuals and process documents for the better understanding of equipment and processes

GS2. document the information from the sketches, engineering drawings, and parameters on various monitoring panels

GS3. communicate effectively with team members

GS4. organize all process/equipment manuals for easy sorting/accessing of information

GS5. detect problems in day-to-day tasks and identify appropriate solutions through discussion with the supervisor

GS6. assist the supervisor in scheduling tasks for the helper and assistant operator

GS7. carry out the assigned tasks within the applicable timelines, maintaining the applicable quality standards

GS8. make work-related decisions, following the authority matrix defined by the organization

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate the electrostatic separators</i>	10	14	-	5
PC1. check the layout and components of electrostatic separators, e.g. drive motor, motor cooling fan, drive shaft and bearings, rollers, lubrication system, power supply unit, etc	3	4	-	2
PC2. conduct electrostatic separation operation maintaining the appropriate parameters, such as load limitation, roll speed, electrode position, voltage, feed material temperature, etc	2	3	-	1
PC3. check and ensure the feed and discharge points are free from obstructions and blockages	3	4	-	1
PC4. follow the appropriate preventive and corrective measures, e.g. earthing, door interlock, cleanup spills, leaks, etc., for the identified hazards	2	3	-	1
<i>Operate the magnetic separator and the screen</i>	16	30	-	12
PC5. use different types of magnetic separators as per the manufacturer's instructions	1	3	-	1
PC6. check for the correct layout and functioning of magnetic separation unit components, e.g. drive mechanism pulley, power supply unit, roller, bearings, splitters, feed arrangements, feed belt, etc.	3	5	-	2
PC7. check for potential issues during the operation of magnetic separator, such as unusual noises, blockage, obstructions, magnetic intensity, leaks, voltage supply, etc	3	5	-	2
PC8. check for blockages at the recommended intervals and clear them appropriately	3	5	-	2
PC9. conduct magnetic separation operation by controlling the feed/ material by adjusting opening of gates, chutes and gaps, adjusting the speed of feeder, and controlling dust by using dust collectors	2	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. remove the excess spillage caused during magnetic separation	1	3	-	1
PC11. carry out processing for separating feed/ material, working around the screens, including control of dust, control of feed/material flow rate	2	4	-	2
PC12. follow the appropriate preventive and corrective measures for safety hazards, such as spillage, abnormal noises/odour, lubrication, leaks, etc	1	2	-	1
<i>Operate the air table</i>	4	6	-	3
PC13. check the moisture of feed material	2	3	-	1
PC14. check the relevant parameters, e.g. inclination of the air table, table cloth, hot air blower, and splitter position	2	3	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4113
NOS Name	Operate the electrostatic separators, magnetic separators, screens, air table
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N4114: Carry out wet table separation, flotation separation, dewatering and screening

Description

This unit is about wet table separation, flotation separation, dewatering and screening the minerals

Scope

The scope covers the following :

- Wet table separation
- Flotation separation
- Dewatering
- Screening

Elements and Performance Criteria

Wet table separation

To be competent, the user/individual on the job must be able to:

- PC1.** check for the correct functioning of wet table separation and drive system, concentrator table, pumping systems, tank, etc
- PC2.** check and record various operating parameters, such as water level, feed rate, pulp density, wash water quantity, etc
- PC3.** check, monitor and record the flow rate and threshold limit values
- PC4.** identify and mitigate the relevant hazards by using ropes/ barriers/ safety signs
- PC5.** check for and report spills and leakages to the supervisor

Flotation separation

To be competent, the user/individual on the job must be able to:

- PC6.** identify different types of flotation
- PC7.** use the appropriate separation equipment, such as pumping system, pressure and water flow control and monitoring system, and valves, following the standard procedure
- PC8.** control the quantity of feed, percent solid, addition of water, level of pulp, launder water, pump speed
- PC9.** identify and mitigate the relevant hazards by using ropes/ barriers/ safety signs
- PC10.** check for and report spills and leakages to the supervisor

Dewatering

To be competent, the user/individual on the job must be able to:

- PC11.** check the slurry pumping system for the correct functioning.
- PC12.** operate different types of dewatering systems, e.g. decanter pumping systems, hydro cyclones, drive systems, feed wells, etc
- PC13.** maintain dewatering systems and components, such as motor and gear box, pumps, slurry pipe lines, valves, etc
- PC14.** check for unusual noises and take appropriate corrective measures

Qualification Pack

PC15. follow the sampling process and use auto sampler as required

PC16. mitigate the identified hazards using rope/ barriers/ safety signs

PC17. identify and report spills and leakages to the supervisor

Screening

To be competent, the user/individual on the job must be able to:

PC18. use different types of screening systems as per the requirement

PC19. check the functioning of screening system, gear box, bearings etc

PC20. conduct screening maintaining the appropriate parameters, e.g. feed rate, quality of over or under size, screen blinding, etc

PC21. monitor the feed bin levels and vibrations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different water purifying and waste treatment agents

KU2. different types of heavy mineral separation processes and associated equipment

KU3. the relevant alarms, pumps, screens, monitor parameter etc

KU4. the use of appropriate reagents as per the process requirements

KU5. different separation systems, such as spiral separation, wet table, concentrators, hindered settling classifiers and components like drive systems etc

KU6. different types of hindered settling classifiers equipment, i.e. pressure and water flow control and monitoring system, valves, pumping system, etc

KU7. different types of dewatering systems, e.g. hydro cyclone, stock tank decanters, pumping systems, sampling systems, rakes, drive systems, feed wells

KU8. the mining safety procedures

KU9. different components of separation equipment and their repair and maintenance

KU10. different types of screening systems and components including feed bin, sampling system, drive system, lubrication system, screen cloth, de-blinding system, vacuum system, gear box, bearings, PLC, SCADA control, pump, hydraulic system, motors, valves, etc

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. document information from the sketches and engineering drawings

GS2. read equipment manuals and process documents for the better understanding of equipment and processes

GS3. discuss task lists, schedules and activities with the supervisor, and assist in scheduling tasks for the subordinates

GS4. communicate effectively with the team members

GS5. plan and organize work for their timely completion

GS6. organize all process/equipment manuals for easy sorting/accessing of information

Qualification Pack

- GS7.** follow the given instructions and work on the areas of improvement
- GS8.** carry out the assigned tasks within the applicable timelines, maintaining the applicable quality standards
- GS9.** make work-related decisions, following the authority matrix defined by the organization

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Wet table separation</i>	7	14	-	5
PC1. check for the correct functioning of wet table separation and drive system, concentrator table, pumping systems, tank, etc	1	3	-	1
PC2. check and record various operating parameters, such as water level, feed rate, pulp density, wash water quantity, etc	2	3	-	1
PC3. check, monitor and record the flow rate and threshold limit values	2	3	-	1
PC4. identify and mitigate the relevant hazards by using ropes/ barriers/ safety signs	1	3	-	1
PC5. check for and report spills and leakages to the supervisor	1	2	-	1
<i>Floatation separation</i>	5	10	-	5
PC6. identify different types of floatation	1	2	-	1
PC7. use the appropriate separation equipment, such as pumping system, pressure and water flow control and monitoring system, and valves, following the standard procedure	1	2	-	1
PC8. control the quantity of feed, percent solid, addition of water, level of pulp, launder water, pump speed	1	2	-	1
PC9. identify and mitigate the relevant hazards by using ropes/ barriers/ safety signs	1	2	-	1
PC10. check for and report spills and leakages to the supervisor	1	2	-	1
<i>Dewatering</i>	10	14	-	6
PC11. check the slurry pumping system for the correct functioning.	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. operate different types of dewatering systems, e.g. decanter pumping systems, hydro cyclones, drive systems, feed wells, etc	1	2	-	1
PC13. maintain dewatering systems and components, such as motor and gear box, pumps, slurry pipe lines, valves, etc	1	2	-	1
PC14. check for unusual noises and take appropriate corrective measures	3	2	-	1
PC15. follow the sampling process and use auto sampler as required	1	2	-	1
PC16. mitigate the identified hazards using rope/ barriers/ safety signs	1	2	-	-
PC17. identify and report spills and leakages to the supervisor	1	2	-	1
<i>Screening</i>	8	12	-	4
PC18. use different types of screening systems as per the requirement	2	2	-	1
PC19. check the functioning of screening system, gear box, bearings etc	3	4	-	1
PC20. conduct screening maintaining the appropriate parameters, e.g. feed rate, quality of over or under size, screen blinding, etc	1	2	-	1
PC21. monitor the feed bin levels and vibrations	2	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4114
NOS Name	Carry out wet table separation, flotation separation, dewatering and screening
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N4115: Carry out general maintenance of ore processing equipment

Description

This unit is about the general maintenance of the equipment used in the mineral separation plant

Scope

The scope covers the following :

- Conduct preventive maintenance of ore processing equipment

Elements and Performance Criteria

Conduct preventive maintenance of ore processing equipment

To be competent, the user/individual on the job must be able to:

- PC1.** follow the manufacturer's instructions concerning the safe handling and maintenance of the machine
- PC2.** carry out greasing/ lubrication of pivot points in the machine
- PC3.** use the appropriate mechanical tools for repairs, following the applicable blueprints and manuals
- PC4.** perform routine inspections of processing equipment, following the manufacturer's instructions
- PC5.** inspect valves, linkages, bearings, hydraulic, and pneumatic systems for smooth operation
- PC6.** carry out troubleshooting, making appropriate adjustments to the equipment and machinery as required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of unit operations relevant to ore processing
- KU2.** the consequences of improper maintenance of ore processing equipment or in operational equipment, e.g. the direct/indirect costs of accidents for the company
- KU3.** the applicable safety regulations
- KU4.** the local emergency response/ disaster management plan
- KU5.** the duties and responsibilities of workmen under Atomic Energy Regulatory Board (AERB) act
- KU6.** the consequence of violating safety procedures
- KU7.** the maintenance of different equipment, such as metering device, pumps, centrifugal pumps, diaphragm pumps, etc.

Generic Skills (GS)

User/individual on the job needs to know how to:



Qualification Pack

- GS1.** follow the online Enterprise Resource Planning (ERP) system under guidance of the supervisor and make note of relevant observations
- GS2.** analyze and interpret drawings, symbols and measurements
- GS3.** read the Instructions given in the Original Equipment Manufacturer (OEM) manuals
- GS4.** discuss task lists, schedules and activities with the supervisor, and assist in scheduling tasks for the subordinates
- GS5.** communicate effectively with co-workers
- GS6.** make appropriate decisions concerning the area of work, applying reasoning skills
- GS7.** plan and organize work for their timely completion
- GS8.** probe to extract appropriate information about the job
- GS9.** follow 5s principles at work
- GS10.** detect problems in day-to-day tasks and identify appropriate solutions through discussion with the supervisor
- GS11.** carry out the assigned tasks within the applicable timelines, maintaining the applicable quality standards

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct preventive maintenance of ore processing equipment</i>	30	50	-	20
PC1. follow the manufacturer's instructions concerning the safe handling and maintenance of the machine	5	4	-	3
PC2. carry out greasing/ lubrication of pivot points in the machine	5	12	-	4
PC3. use the appropriate mechanical tools for repairs, following the applicable blueprints and manuals	5	5	-	3
PC4. perform routine inspections of processing equipment, following the manufacturer's instructions	5	12	-	3
PC5. inspect valves, linkages, bearings, hydraulic, and pneumatic systems for smooth operation	5	5	-	3
PC6. carry out troubleshooting, making appropriate adjustments to the equipment and machinery as required	5	12	-	4
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4115
NOS Name	Carry out general maintenance of ore processing equipment
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N4116: Perform water treatment and management of subgrade ore

Description

This unit is about water treatment and management of subgrade ore.

Scope

The scope covers the following :

- Treat water
- Manage the subgrade ore

Elements and Performance Criteria

Treat water

To be competent, the user/individual on the job must be able to:

- PC1.** deliver the processed water to the settling area
- PC2.** test the sample water and relevant parameters, and monitor the pond levels
- PC3.** adjust the pH of the ore processing plant wastewater by adding alkaline substances, such as lime or sodium hydroxide, removing metals and sulfates from the ore to restore the optimal pH levels of water
- PC4.** carry out coagulation and sedimentation for water treatment, to remove the fine particles and suspended solids from the water
- PC5.** carry out oxidation by adding oxidizing agents, such as sodium hypochlorite or ozone, to degrade the residual organic compounds and metals in the water
- PC6.** perform water filtration by passing the water through a filter media, such as sand, activated carbon, or membrane to remove any remaining particles or impurities
- PC7.** recycle the treated water in mining operations, such as flotation, washing, or cooling, to reduce water consumption and associated costs

Manage the subgrade ore

To be competent, the user/individual on the job must be able to:

- PC8.** carry out appropriate beneficiation activities, such as crushing, grinding, screening, washing, magnetic separation, flotation, etc., to remove impurities and increase the concentration of the desired mineral
- PC9.** carry out blending of subgrade ore, mixing it with high grade ore to achieve the desired grade for the final product
- PC10.** stockpile the subgrade ore in the designated area for future use or disposal, performing its effective management to avoid environmental impacts

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the procedure of testing the sample water and relevant parameters

Qualification Pack

- KU2.** how to adjust the pH of the ore processing plant wastewater
- KU3.** how to carry out coagulation and sedimentation for water treatment
- KU4.** how to carry out oxidation of water by adding appropriate oxidizing agents
- KU5.** the water filtration procedure using the appropriate filter media, such as sand, activated carbon, or membrane
- KU6.** the recycling of the treated water in different mining operations, e.g. flotation, washing, or cooling, to reduce water consumption and associated costs
- KU7.** the characteristics of subgrade ore and its appropriate management
- KU8.** the environmental concerns associated with the handling of subgrade ore
- KU9.** how to carry out appropriate beneficiation activities, such as crushing, grinding, screening, washing, magnetic separation, flotation, etc
- KU10.** the blending of subgrade ore, mixing it with high grade ore to achieve the desired grade
- KU11.** the appropriate precautions for stockpiling the subgrade ore

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** document information from the sketches and engineering drawings
- GS2.** enter the relevant information in the online ERP system under the guidance of the supervisor
- GS3.** read equipment manuals and process documents for the better understanding of equipment and processes
- GS4.** discuss task lists, schedules and activities with the supervisor
- GS5.** communicate effectively with the team members
- GS6.** plan and organize work for its timely completion
- GS7.** follow the given instructions and work on the areas of improvement
- GS8.** carry out the assigned tasks within the applicable timelines, maintaining the applicable quality standards
- GS9.** make work-related decisions, following the authority matrix defined by the organization

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Treat water</i>	20	35	-	15
PC1. deliver the processed water to the settling area	2	3	-	2
PC2. test the sample water and relevant parameters, and monitor the pond levels	4	6	-	3
PC3. adjust the pH of the ore processing plant wastewater by adding alkaline substances, such as lime or sodium hydroxide, removing metals and sulfates from the ore to restore the optima pH levels of water	2	5	-	2
PC4. carry out coagulation and sedimentation for water treatment, to remove the fine particles and suspended solids from the water	4	6	-	3
PC5. carry out oxidation by adding oxidizing agents, such as sodium hypochlorite or ozone, to degrade the residual organic compounds and metals in the water	4	6	-	3
PC6. perform water filtration by passing the water through a filter media, such as sand, activated carbon, or membrane to remove any remaining particles or impurities	2	6	-	1
PC7. recycle the treated water in mining operations, such as flotation, washing, or cooling, to reduce water consumption and associated costs	2	3	-	1
<i>Manage the subgrade ore</i>	10	15	-	5
PC8. carry out appropriate beneficiation activities, such as crushing, grinding, screening, washing, magnetic separation, flotation, etc., to remove impurities and increase the concentration of the desired mineral	4	5	-	2
PC9. carry out blending of subgrade ore, mixing it with high grade ore to achieve the desired grade for the final product	4	5	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. stockpile the subgrade ore in the designated area for future use or disposal, performing its effective management to avoid environmental impacts	2	5	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4116
NOS Name	Perform water treatment and management of subgrade ore
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1810: Follow the applicable health, safety and environmental guidelines (Placer Mining)

Description

This unit is about following the applicable health, safety and environmental guidelines.

Scope

The scope covers the following :

- Maintain personal health and safety
- Assist in hazard management
- Work on equipment and machinery
- Check the first aid box, firefighting and safety equipment
- Report incidents

Elements and Performance Criteria

Maintain personal health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers
- PC2.** ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments
- PC3.** clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes
- PC4.** follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC5.** follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking
- PC6.** use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose
- PC7.** identify six directional hazards at the workplace and take decisions accordingly
- PC8.** follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas
- PC9.** undertake the Take-5 (Personal Risk Assessment) before the commencement of any work
- PC10.** lift objects carefully maintaining the correct body posture to avoid injury
- PC11.** use self-rescue apparatus appropriately when required
- PC12.** work in the assigned work zone in the designated work area
- PC13.** follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc
- PC14.** wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant
- PC15.** use caution signs as applicable and obey safety signs, stickers, and tags

Qualification Pack

PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact

PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help

Assist in hazard management

To be competent, the user/individual on the job must be able to:

PC18. label and store chemicals appropriately

PC19. display the MSDS for the hazardous chemicals in storage and handling areas

PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines

PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre

PC22. maintain regular and open communication with co-workers and the employer

Work on equipment and machinery

To be competent, the user/individual on the job must be able to:

PC23. check and ensure every instrument, apparatus and equipment is AERB approved

PC24. operate the appropriate tools, equipment and machinery with proper training and authorization

Check the first aid box, firefighting, and safety equipment

To be competent, the user/individual on the job must be able to:

PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible

PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas

PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors

PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite

Report incidents

To be competent, the user/individual on the job must be able to:

PC29. identify and report potential hazards/ unsafe conditions and practices

PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.

PC31. report the emergencies and any safety incidents as per the organizational procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of chemical plants and their operations

KU2. the importance of time-keeping, discipline and punctuality

KU3. various chemical processes carried out in a rare earth extraction plant

KU4. the impact of violating applicable safety procedures

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- KU5.** the relevant quality norms
- KU6.** different types of individual rare earth chemical processes and the use of associated equipment
- KU7.** different types of effluent treatment processes and the use of associated equipment
- KU8.** the use of different water purifying and waste treatment agents
- KU9.** the hazards and the appropriate safety measures in acid/ alkali handling activities and usage of relevant PPE
- KU10.** hazards and safety aspects involved in acid/ alkali handling activities and usage of relevant PPE
- KU11.** the internal safety organization and the role of the safety committee, workforce inspector and AERB, etc
- KU12.** the importance of sensitization towards different genders and persons with disabilities (PWD)
- KU13.** the Importance of following infection control policies, '5-S' practices, and waste management as prescribed by regulatory authorities like the Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.
- KU14.** the importance of water/material/energy conservation and management
- KU15.** the importance of Social Management Plan (SMP) and Environment Management Plan (EMP) prepared by the organization, as per DGMS guideline
- KU16.** the importance of wearing own TLD or pocket dosimeter (to be obtained from the Health Physics Unit) in the active area
- KU17.** the importance of staying alert and attentive to the work surroundings
- KU18.** the importance of staying in the assigned work zone in the designated work area
- KU19.** the benefits of making the safety rules part of new employee orientation
- KU20.** the importance of contacting the supervisor or manager for instruction, guidance, or training when required

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal health and safety</i>	16	24	-	11
PC1. follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers	1	2	-	1
PC2. ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments	1	1	-	1
PC3. clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes	1	1	-	-
PC4. follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	-
PC5. follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking	1	2	-	-
PC6. use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose	1	2	-	-
PC7. identify six directional hazards at the workplace and take decisions accordingly	1	2	-	1
PC8. follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas	1	2	-	1
PC9. undertake the Take-5 (Personal Risk Assessment) before the commencement of any work	1	2	-	1
PC10. lift objects carefully maintaining the correct body posture to avoid injury	-	1	-	-
PC11. use self-rescue apparatus appropriately when required	1	1	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. work in the assigned work zone in the designated work area	1	1	-	1
PC13. follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc	1	1	-	1
PC14. wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant	1	1	-	1
PC15. use caution signs as applicable and obey safety signs, stickers, and tags	1	1	-	-
PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact	1	1	-	1
PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help	1	1	-	1
<i>Assist in hazard management</i>	5	8	-	4
PC18. label and store chemicals appropriately	1	1	-	1
PC19. display the MSDS for the hazardous chemicals in storage and handling areas	1	2	-	1
PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines	1	1	-	-
PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre	1	2	-	1
PC22. maintain regular and open communication with co-workers and the employer	1	2	-	1
<i>Work on equipment and machinery</i>	2	4	-	1
PC23. check and ensure every instrument, apparatus and equipment is AERB approved	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. operate the appropriate tools, equipment and machinery with proper training and authorization	1	2	-	1
<i>Check the first aid box, firefighting, and safety equipment</i>	4	10	-	2
PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible	1	3	-	-
PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas	1	2	-	-
PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors	1	2	-	1
PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite	1	3	-	1
<i>Report incidents</i>	3	4	-	2
PC29. identify and report potential hazards/ unsafe conditions and practices	1	2	-	-
PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.	1	1	-	1
PC31. report the emergencies and any safety incidents as per the organizational procedure	1	1	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1810
NOS Name	Follow the applicable health, safety and environmental guidelines (Placer Mining)
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1811: Work collaboratively and effectively in a team

Description

This unit is about the importance of teamwork in the workplace.

Scope

The scope covers the following :

- Work effectively with co-workers
- Communicate effectively with co-workers
- Practice inclusiveness at work

Elements and Performance Criteria

Work effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC1.** identify own direct and indirect responsibilities, e.g. duties concerning team members
- PC2.** undertake appropriate activities to promote cooperation and good relationships with co-workers
- PC3.** mentor and assist subordinates in the execution of their work responsibilities
- PC4.** identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures
- PC5.** follow the recommended practices to avoid and resolve conflicts at work
- PC6.** interact effectively and cooperate with co-workers
- PC7.** manage workplace conflicts, ensuring no impact on work
- PC8.** identify and take responsibility for own mistakes
- PC9.** follow the organizational policies and procedures
- PC10.** follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.
- PC11.** exchange feedback with peers and other team members to improve the quality of work

Communicate effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC12.** follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel
- PC13.** communicate clearly and politely to ensure effective communication with co-workers
- PC14.** practice active listening to understand the conveyed information and instructions
- PC15.** practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication
- PC16.** maintain the professional standard of communication, e.g. avoiding the use of offensive language at work

Practice inclusiveness at work

To be competent, the user/individual on the job must be able to:

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- PC17.** empathize with Persons with Disabilities (PwD)
- PC18.** practice gender equality at work
- PC19.** maintain respectful behaviour and etiquette at work
- PC20.** listen to and acknowledge the feelings, concerns, opinions, and ideas of others

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of teamwork and the consequence of poor teamwork on project outcomes, timelines, safety, etc
- KU2.** the duties and rights of workers
- KU3.** the importance of creating a healthy and cooperative work environment
- KU4.** the importance of proper and effective communication and the adverse effects of poor communication on quality, timeliness, safety, etc
- KU5.** different modes of communication used at work
- KU6.** the importance and need of supporting co-workers facing problems for the smooth functioning of work
- KU7.** the fundamental concept of gender equality
- KU8.** how to recognize and be sensitive to issues of disability, culture and gender
- KU9.** how to show empathy to PwD appropriately
- KU10.** legislation, policies, and procedures relating to gender sensitivity and cultural diversity including their impact on the area of operation
- KU11.** the appropriate behaviour to be practised with coworkers
- KU12.** the appropriate communication skills and leadership qualities
- KU13.** how to be open to listening to and understanding the other person's point of view

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with co-workers</i>	15	30	-	10
PC1. identify own direct and indirect responsibilities, e.g. duties concerning team members	3	5	-	3
PC2. undertake appropriate activities to promote cooperation and good relationships with co-workers	1	2	-	1
PC3. mentor and assist subordinates in the execution of their work responsibilities	1	2	-	1
PC4. identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures	3	4	-	2
PC5. follow the recommended practices to avoid and resolve conflicts at work	1	2	-	1
PC6. interact effectively and cooperate with co-workers	1	1	-	-
PC7. manage workplace conflicts, ensuring no impact on work	1	1	-	-
PC8. identify and take responsibility for own mistakes	1	3	-	1
PC9. follow the organizational policies and procedures	1	3	-	-
PC10. follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.	1	4	-	1
PC11. exchange feedback with peers and other team members to improve the quality of work	1	3	-	-
<i>Communicate effectively with co-workers</i>	9	10	-	5
PC12. follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel	2	3	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. communicate clearly and politely to ensure effective communication with co-workers	1	2	-	-
PC14. practice active listening to understand the conveyed information and instructions	2	1	-	1
PC15. practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication	2	2	-	1
PC16. maintain the professional standard of communication, e.g. avoiding the use of offensive language at work	2	2	-	1
<i>Practice inclusiveness at work</i>	6	10	-	5
PC17. empathize with Persons with Disabilities (PwD)	2	2	-	2
PC18. practice gender equality at work	2	2	-	2
PC19. maintain respectful behaviour and etiquette at work	1	3	-	1
PC20. listen to and acknowledge the feelings, concerns, opinions, and ideas of others	1	3	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1811
NOS Name	Work collaboratively and effectively in a team
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N4109.Carry out rare earth material ore processing	30	50	0	20	100	15
MIN/N4110.Convey the material from mine for processing	30	50	0	20	100	10
MIN/N4111.Operate the spiral separator and density separator	30	50	0	20	100	10
MIN/N4112.Operate the drier and blower	30	50	0	20	100	10
MIN/N4113.Operate the electrostatic separators, magnetic separators, screens, air table	30	50	0	20	100	10
MIN/N4114.Carry out wet table separation, flotation separation, dewatering and screening	30	50	0	20	100	10
MIN/N4115.Carry out general maintenance of ore processing equipment	30	50	0	20	100	10
MIN/N4116.Perform water treatment and management of subgrade ore	30	50	0	20	100	10
MIN/N1810.Follow the applicable health, safety and environmental guidelines (Placer Mining)	30	50	0	20	100	5



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1811.Work collaboratively and effectively in a team	30	50	0	20	100	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	320	530	-	200	1050	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.